

Power Architecture Derivations

Relative Institutional Packages and Type-Dependent Participation

Working formal appendix

2026-05-11

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1 Status

This is the clean working surface for the reimplemented model architecture. The prior feasibility-branch derivation has been archived in git tag `redesign-feasibility-branch-2026-05-11`. Do not import theorem statements, notation, or branch labels from that archived architecture without rederiving them here.

The active manuscript `formal_model_v5.Rmd` remains frozen. New formal work should be developed in this file first, checked by scripts, independently verified, and only then transported to the paper.

1.1 Architecture Decision

The main baseline no longer uses fixed transfers that may fit one realized pie but not another. Proposals are relative institutional packages: quota rules, production shares, cut obligations, enforcement provisions, exceptions, and flexibility terms. These packages are feasible in every state.

Screening comes from H's type-dependent participation threshold:

$$U_H(y, \theta) = y + b_H(\theta),$$

where y is the concession offered by weak states and $b_H(\theta)$ is the direct benefit that H obtains from being inside the agreement. H accepts if the package payoff exceeds its continuation value.

2 Q&A

What is the substantive puzzle?

Why would a powerful hegemon choose unanimity or consensus when that rule appears to constrain its formal power?

What mechanism should the model isolate?

Unanimity can make an informed hegemon pivotal. Weak states must offer a package that satisfies H's participation constraint, but they do not observe how favorable the package must be for H to prefer agreement to its outside option.

What is the key modeling reset?

The proposal is a relative institutional package, not an absolute transfer. It is always feasible. The private-information problem is about H's type-specific acceptance threshold.

What is the OPEC interpretation?

Members bargain over quotas, cuts, shares, monitoring, exceptions, and flexibility. They can observe the proposed package, but they do not know Saudi Arabia's true participation threshold because Saudi Arabia privately knows its spare capacity, opportunity cost, and outside option.

3 Primitives

There are $N \geq 3$ states: one hegemon H and $m = N - 1$ weak states. Nature draws

$$\theta \in \{0, 1\}.$$

The hegemon observes θ . Weak states share prior belief

$$\mu = \Pr(\theta = 1).$$

The institutional value is

$$V(0) = 1, \quad V(1) = r, \quad r > 1,$$

and

$$V_e(\mu) = 1 + \mu(r - 1).$$

The discount factor is $\beta \in (0, 1]$.

Agenda power is represented by

$$\pi_H \in [0, 1].$$

The baseline sets

$$\pi_H = 0,$$

so only weak states are recognized as proposers. This isolates informational power through pivotality from proposal power.

4 Relative Package

A proposal is a package

$$x \in X.$$

For the first reimplementation, use a scalar concession index

$$y \in [0, \bar{y}],$$

where a higher y is more favorable to H and less favorable to the weak coalition. The package is feasible if $y \in [0, \bar{y}]$. Feasibility does not depend on θ .

Substantively, y may represent a larger production share for H , a weaker cut obligation, more flexibility, better enforcement against other members, or more favorable treatment under changing market conditions.

4.1 H Payoff

The minimal payoff representation is

$$U_H(y, \theta) = y + b_H(\theta). \tag{H-payoff}$$

The term $b_H(\theta)$ is the direct benefit H obtains from the agreement itself, apart from the concession y . It may capture cartel coordination, price stability, predictability, reputation, or the value of constraining other members.

Let

$$C_H(\theta, \mu')$$

be H 's continuation value after rejection, given the posterior μ' . H accepts package y in state θ iff

$$y + b_H(\theta) \geq \beta C_H(\theta, \mu'). \quad (\text{H-PC})$$

Define the type-specific acceptance threshold

$$y_\theta^*(\mu') = \beta C_H(\theta, \mu') - b_H(\theta). \quad (\text{threshold})$$

The screening condition is

$$y_1^*(\mu') > y_0^*(\mu'). \quad (\text{screening})$$

Equivalently, in a one-shot participation version with outside option $o_H(\theta)$,

$$y_\theta^* = o_H(\theta) - b_H(\theta),$$

so screening requires

$$o_H(1) - b_H(1) > o_H(0) - b_H(0).$$

The high type requires a larger concession because its outside option net of the direct benefit of agreement is higher.

4.2 Weak-State Payoffs

The next derivation must specify how y reduces the payoff available to weak states. The intended normalization is one-for-one in reduced form: increasing the concession to H by one unit reduces the surplus available to the weak coalition by one unit.

If a weak proposer needs a coalition of weak voters to accept, each non-proposing weak voter receives its continuation value. The proposer keeps the residual. This preserves the Baron-Ferejohn logic without using state-contingent feasibility failure.

This object is still pending formalization.

5 Baseline Timing: $\pi_H = 0$

The first proof pass keeps $\pi_H = 0$ in both rounds.

1. Nature draws θ ; H observes θ , weak states observe only μ .
2. A weak state is recognized as proposer.
3. The proposer offers a relative institutional package y .
4. Under unanimity, H must accept the package for agreement to form.
5. Weak voters receive continuation-value offers required for coalition approval.
6. If agreement fails, the game proceeds to the next bargaining round with beliefs updated by Bayes' rule on path.

No branch in this baseline relies on a package being feasible in one state and infeasible in another.

6 Objects To Derive

6.1 R2 Unanimity

Derive the terminal-round weak-proposer problem under unanimity. Candidate packages:

1. **Pooling package:** offer y_1^* , both types of H accept.
2. **Low-only package:** offer y_0^* , low H accepts and high H rejects.
3. **No-agreement package:** induce rejection by both types if this is payoff-relevant under the stated protocol.

The derivation must produce:

$$W_2^U(\mu), \quad C_H(0, \mu), \quad C_H(1, \mu).$$

6.2 R1 Unanimity

Using the R2 continuation values, derive the Round-1 weak-proposer problem under unanimity. Candidate branches:

1. pooling/high-threshold package;
2. low-only package;
3. no-information rejection or continuation, if implementable under the stated protocol.

The proof must verify sequential rationality and Bayesian consistency.

6.3 Majority

Recompute majority under $\pi_H = 0$. The expected baseline is that majority generates no screening when weak states can exclude H and H receives its outside option externally. This must be rederived in the new notation.

6.4 Entry and Institutional Choice

After R2, R1, and majority are rederived, rebuild:

1. weak-state entry and nesting;
2. conditional institutional comparison for H;
3. calibrated OPEC classification.

7 Reproducibility Plan

Create new scripts rather than modifying the archived C-B-R scripts in place. Planned scripts:

1. `scripts/verify_relative_package_R2_piH0.R`
2. `scripts/verify_relative_package_R1_piH0.R`
3. `scripts/verify_relative_package_majority_piH0.R`
4. `scripts/verify_relative_package_entry_nesting_piH0.R`
5. `scripts/verify_relative_package_classification_piH0.R`

Each result should be independently verified before being promoted in the proof ledger.

8 Proof Ledger

Object	Status	Notes
Archived feasibility-branch derivation	Archived	Preserved in tag redesign-feasibility-branch-2026-05-11 ; do not use as current architecture.
Relative-package architecture	Adopted	Packages are feasible in every state; screening comes from type-dependent participation thresholds.
Minimal H payoff primitive	Adopted	$U_H(y, \theta) = y + b_H(\theta)$.
Screening condition	Adopted as target primitive	Need $y_1^*(\mu') > y_0^*(\mu')$, equivalently $\beta C_H(1, \mu') - b_H(1) > \beta C_H(0, \mu') - b_H(0)$.
Weak-state cost of concession	Pending formalization	Intended reduced-form normalization: y is costly one-for-one to the weak coalition.
R2 unanimity under relative packages	Pending	First derivation target.
R1 unanimity under relative packages	Pending	Derive only after R2 is closed.
Majority under relative packages	Pending	Must keep H's outside option external to the institutional pie.
Entry/nesting	Pending	Rebuild after R1 and majority.
Institutional classification	Pending	Rebuild after entry and conditional comparison.
$\pi_H > 0$ extensions	Pending	Treat agenda power separately after the baseline is verified.